Lakshay 100002552 6 Sec-M

Date
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Assignment - I

Q1) Database ⇒ Organized collection of data, where information is stored in a structured way.

* DBMS ⇒ It is a software that helps you create, manage and control databases.
Examples ⇒ MySQL, oracle, MongoDB etc.
It can store and retrieve data easily and quickly. It allows multiple users, provides security. It also gives backup and recovery.

Q2) * DDL commands -
1) CREATE ⇒ Creation of a new table/database.
2) ALTER ⇒ Modify an existing table.
3) DROP ⇒ Drop table/database permanently
4) TRUNCATE ⇒ Removes all records from table.

* DML commands

1) SELECT ⇒ Retrives data
2) INSERT ⇒ Add new data
3) UPDATE ⇒ Modify existing data
4) DELETE ⇒ Removes data

Q3) Problems with FSDM -
1) Data Redundancy ⇒ Duplicate data
2) Data Inconsistency
3) Data Isolation
4) Difficult to access
5) Integrity problems
6) security concerns
7) No backup or recovery

Q4) View of data ⇒ The way data is seen by different users, even though the actual data is stored in a complex way inside the database. It has 3 levels.
1) Physical Level ⇒ How data is actually stored in memory.
- Example ⇒ Student data stored in disk file.
2) Logical Level ⇒ What data is stored.
- Eg ⇒ STUDENT(ID, Name, Age, Course)

3) View Level ⇒ What user can actually see.

- Example ⇒ Teacher's View ⇒ ID, Name, Marks etc.
  Student's View ⇒ Name, Course etc.

Q5) RDMS ⇒ Is a DBMS where data is stored in tabular form i.e rows and column no.

- Components

1) Table ⇒ Data is stored in table.
2) Tuple ⇒ Each row in a table.
3) Attribute ⇒ Each column of a table
4) Domain ⇒ Set of valid values
5) Schema ⇒ Structure of database.
6) Keys ⇒ Used to idenity and connect data.